



Field and Wave Electromagnetics

电磁场与电磁波

Chapter 6: Static magnetic fields

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Chapter Title		课时	
1	Static Electric Fields	3	静电场： (1) 期末<15%，不考大题； (2) 磁矢等难度大尽量简化
2	Solution of Electrostatic Problems		
4	Steady Electric Currents	2	
5	Basic Static Magnetic Fields	2	
7	Time-varying Fields and Maxwell's Equations	5	电压计算等适当简化，
9	Plane Electromagnetic Waves	11	
13	Waveguides and Cavity Resonators	11	重点强化章节，期末有一道推导大题
16	Antennas and Radiating systems	3	
	机动/答疑	2	
		40	

1 总体原则：弱化电路相关(V,A,I,U)，强化Photonics相关(E,H)，突出本课程对光电专业支撑作用

2 删掉“传输线”，增加“天线”；删除“镜像法”（分离变量法保留）

3 课时可以根据情况适当调整，预留2课时为机动/答疑课时；

4 平时成绩40分，（作业上交10分+点名10分，作业完成情况20分）

5 作业完成情况20分学生两种选择（默认选择a）：

（a）作业上交点名20分+作业完成情况(绑定期末作业题)20分

（b）助教随机抽出4个作业大题，每题5分，1小时完成(答疑课时候完成)，得分视为平时成绩。

Chapter 3: Static Electric Fields

Field: A spatial distribution of a scalar or vector quantity which may or may not be a function of time.

1. Fundamental postulates of electrostatic in free space

2. Coulomb's law

3. Gauss's law

4. Conductors in static electric field

5. Dielectrics in static electric field

6. Electric flux density

7. Boundary conditions for electrostatic fields

8. Capacitance and capacitors

9. Electrostatic energy and forces

$$\nabla \cdot \vec{E} = \frac{\rho}{\epsilon_0} \quad \oint_S \vec{E} \cdot d\vec{s} = \frac{Q}{\epsilon_0}$$

$$\nabla \times \vec{E} = 0 \quad \oint_C \vec{E} \cdot d\vec{l} = 0$$

$$\vec{D} = \epsilon_0 \vec{E} + \vec{P}$$

$$\nabla \cdot \vec{D} = \rho$$

$$\oint_S \vec{D} \cdot d\vec{s} = Q$$

Chapter 4: Solutions of Electrostatic Problems

4-1 Introduction

4-2 Poisson's and Laplace's Equations $\nabla^2 V = -\frac{\rho}{\epsilon}$ $\nabla^2 V = 0$

4-3 Uniqueness of Electrostatic Solutions

4-4 Methods of Images

4-5 Boundary-Value Problems in Cartesian Coordinates

4-6 Boundary-Value Problems in Cylindrical Coordinates

4-7 Boundary-Value Problems in Spherical Coordinates

Chapter 5: Steady Electric Currents

5-1 Introduction

5-2 Current density and Ohm's law

5-3 Electromotive force and Kirchhoff's voltage law

5-4 Continuity of equation and Kirchhoff's current law

5-5 Power dissipation and Joule's law

5-6 Boundary conditions for current density

5-7 Resistance calculation

Current density

$$\mathbf{J} = Nq\mathbf{u}$$

$$\mathbf{J} = \rho\mathbf{u} \quad (\text{A/m}^2),$$

Continuity of equation

$$\nabla \cdot \mathbf{J} = -\frac{\partial \rho}{\partial t} \quad (\text{A/m}^3).$$

Ohm's law

$$\mathbf{J} = \sigma\mathbf{E} \quad (\text{A/m}^2),$$

Joule's law

$$p = \vec{J} \cdot \vec{E} = \frac{J^2}{\sigma} = \sigma E^2$$

Boundary conditions for current density

Governing Equations for Steady Current Density	
Differential Form	Integral Form
$\nabla \cdot \mathbf{J} = 0$	$\oint_S \mathbf{J} \cdot d\mathbf{s} = 0$
$\nabla \times \left(\frac{\mathbf{J}}{\sigma}\right) = 0$	$\oint_C \frac{1}{\sigma} \mathbf{J} \cdot d\boldsymbol{\ell} = 0$

$$J_{1n} = J_{2n}$$

$$\frac{J_{1t}}{J_{2t}} = \frac{\sigma_1}{\sigma_2}$$

5-7 Resistance calculations

Conductor resistance

$$R = \frac{V}{I} = \frac{-\oint_L \vec{E} \times d\vec{l}}{\oint_S \vec{J} \times d\vec{S}} = \frac{-\oint_L \vec{E} \times d\vec{l}}{\oint_S \sigma \vec{E} \times d\vec{S}}$$

The procedure for computing the resistance of a piece of conducting material between specified equipotential surfaces (or terminals):

1. Choose an appropriate coordinate system.
2. Assume a potential difference V_0 between conductor terminals.
3. Find electric field intensity $\vec{E}$ within the conductor.

homogeneous material: Laplace's equation $\nabla^2 V = 0$ $\vec{E} = -\nabla V$

4. Find the total current

$$I = \oint_S \vec{J} \times d\vec{S} = \oint_S \sigma \vec{E} \times d\vec{S}$$

5. Find resistance R by taking the ratio V_0/I .

Note that, for this problem, it is not convenient to begin by assuming a total current I because it is not obvious how $\vec{E}$ varies with r for a given I . Without $\vec{E}$ and $\vec{J}$ cannot be determined.

Chapter 6: Static magnetic fields

6-1 Introduction

6-2 Fundamental postulates of magnetostatics in free space

6-3 Vector magnetic potential

6-4 The Biot-Savart law and applications

6-5 The magnetic dipole

6-6 Magnetization and equivalent current densities

6-7 Magnetic field intensity and relative permeability

6-8 Magnetic circuits

6-9 Behavior of magnetic materials

6-10 Boundary conditions for magnetostatic fields

6-11 Inductances and inductors

6-12 Magnetic energy

6-13 Magnetic forces and torques

6-1 Introduction

□ Electric force: $\vec{F} = q\vec{E} \quad (N)$

□ Magnetic force: $\vec{F}_m = q\vec{u}' \times \vec{B} \quad (N)$

□ Electromagnetic force: $\vec{F} = q(\vec{E} + \vec{u}' \times \vec{B}) \quad (N)$

Lorentz's force equation

6.2 Fundamental Postulates of Magnetostatics in Free Space

Free space

- Static Electric Field

$$\vec{\nabla} \cdot \vec{E} = \rho / \epsilon_0$$

$$\vec{\nabla} \times \vec{E} = 0$$

- a) Electric field comes from electric charges (static);
- b) Magnetic field comes from motion of charges;
- c) *A vector quantity is uniquely specified once its divergence and curl are specified.*

- Static Magnetic Field

$$\vec{\nabla} \cdot \vec{B} = 0$$

$$\vec{\nabla} \times \vec{B} = \mu_0 \vec{J}$$

Steady current

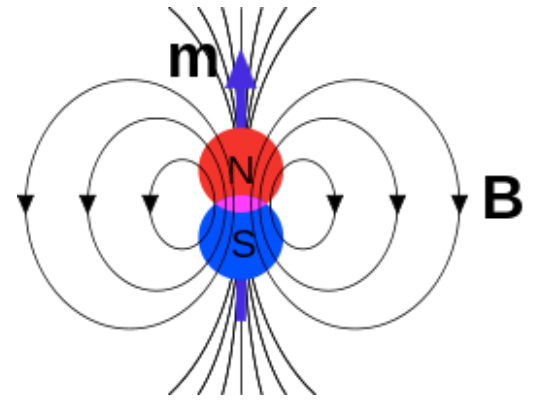
$$\vec{\nabla} \cdot \vec{J} = 0$$

Permeability of free space

$$\mu_0 = 4\pi \times 10^{-7} \text{ Hm}^{-1}$$

$$\vec{\nabla} \cdot \vec{B} = 0 \Rightarrow \oint_s \vec{B} \cdot d\vec{s} = 0$$

- No magnetic flow sources
- Magnetic flux lines always close
- ***Law of conservation of magnetic flux***
- Each magnets has a north and south poles
- Magnetic poles cannot be isolated



(a)



(b)



(c)

$$\vec{\nabla} \times \vec{B} = \mu_0 \vec{J}$$



$$\int_s (\vec{\nabla} \times \vec{B}) \cdot d\vec{s} = \mu_0 \int_s \vec{J} \cdot d\vec{s}$$



Ampere's circuital law

$$\oint_c \vec{B} \cdot d\vec{\ell} = \mu_0 I$$

which states that *the circulation of the magnetic flux density in free space around any closed path is equal to μ_0 times the total current flowing through the surface bounded by the path.*

Postulates of Magnetostatics in Free Space

Differential Form

Integral Form

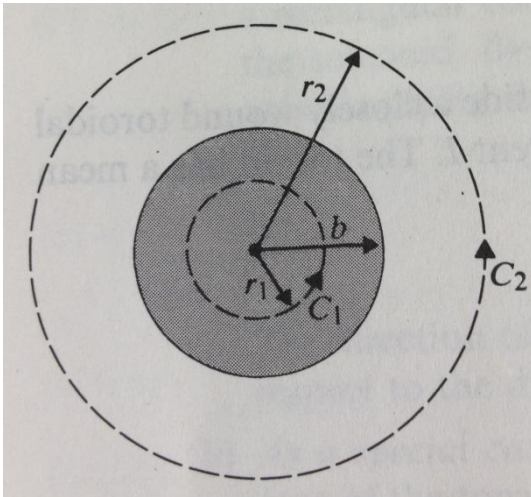
$$\nabla \cdot \mathbf{B} = 0$$

$$\oint_S \mathbf{B} \cdot d\mathbf{s} = 0$$

$$\nabla \times \mathbf{B} = \mu_0 \mathbf{J}$$

$$\oint_C \mathbf{B} \cdot d\boldsymbol{\ell} = \mu_0 I$$

Q6-1 An infinitely long, straight conductor with a circular cross section of radius b carries a steady current I . Determine the magnetic flux density both inside and outside the conductor.



(1) Inside the conductor

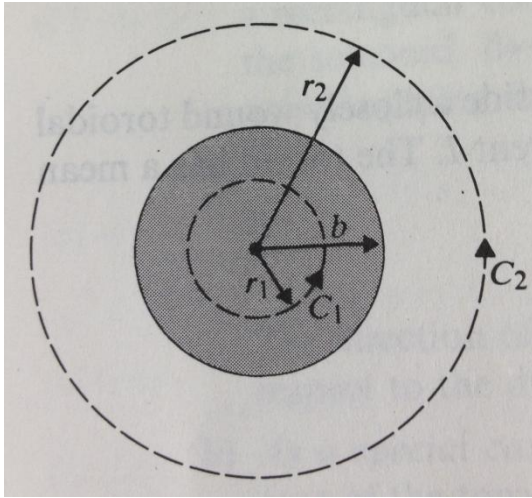
$$\vec{B}_1 = \vec{a}_f B_{f1}$$

$$d\vec{\ell} = \vec{a}_f r_1 df$$

$$\oint_{C_1} \vec{B}_1 \times d\vec{\ell} = \oint_0^{2\pi} B_{f1} r_1 df = 2\pi r_1 B_{f1}$$

The current through the area enclosed by C_1 is $I_1 = \frac{\pi r_1^2}{\pi b^2} I = \left(\frac{r_1}{b}\right)^2 I$

From Ampere's circuital law $\oint_C \vec{B} \times d\vec{\ell} = \mu_0 I$ $\vec{B}_1 = \vec{a}_f B_{f1} = \vec{a}_f \frac{\mu_0 r_1 I}{2\pi b^2}, \quad r_1 \leq b.$



b) Outside the conductor:

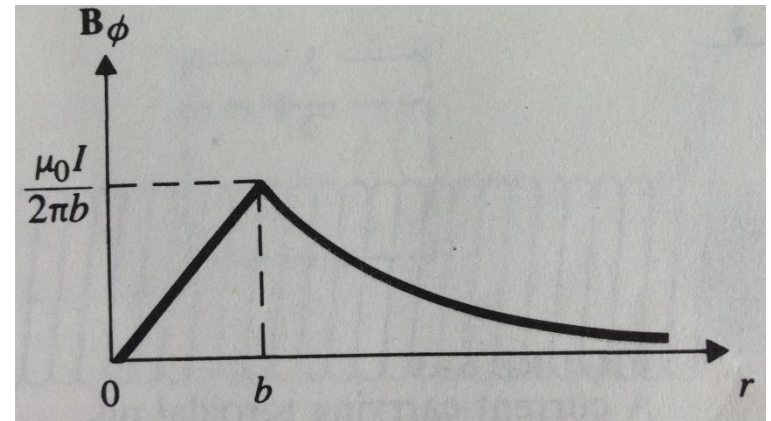
$$\vec{B}_2 = \vec{a}_f B_{f2} \quad d\vec{\ell} = \vec{a}_f r_2 df$$

$$\oint_{C_2} \vec{B}_2 \times d\vec{\ell} = 2\pi r_2 B_{f2}$$

The current through the area enclosed by C_2 is I

$$\vec{B}_2 = \vec{a}_f B_{f2} = \vec{a}_f \frac{\mu_0 I}{2\pi r_2}, \quad r_2 \geq b.$$

The magnitude of B increases linearly with r_1 from 0 until $r_1=b$, after which it decreased inversely with r_2 .



6-3 Vector Magnetic Potential

- Vector magnetic potential (矢量磁势), $\vec{A}$:

$$\boxed{\vec{B} = \vec{\nabla} \times \vec{A}} \leftarrow \boxed{\vec{\nabla} \cdot \vec{B} = 0} \quad (\text{divergence-free, solenoidal, 无散度的})$$

$$c.f.: \vec{E} = -\vec{\nabla} f \Leftarrow \boxed{\vec{\nabla} \times \vec{E} = 0}$$

- Vector Poisson's equation (矢量泊松方程)

$$\vec{\nabla} \times (\vec{\nabla} \times \vec{A}) = \vec{\nabla}(\vec{\nabla} \cdot \vec{A}) - \vec{\nabla}^2 \vec{A} = m_0 \vec{J} \leftarrow \boxed{\vec{\nabla} \times \vec{B} = m_0 \vec{J}}$$

$$\text{P} \quad \boxed{\vec{\nabla} \cdot \vec{A} = 0}$$

Coulomb gauge
(库伦规范)

$$\boxed{\vec{\nabla}^2 \vec{A} = -m_0 \vec{J}}$$

*Vector Poisson's
equation*

In Cartesian coordinates, we have

$$\boxed{\vec{\nabla}^2 \vec{A} = -\mu_0 \vec{J}} \Rightarrow \begin{cases} \vec{\nabla}^2 A_x = -\mu_0 J_x \\ \vec{\nabla}^2 A_y = -\mu_0 J_y \\ \vec{\nabla}^2 A_z = -\mu_0 J_z \end{cases} \xrightarrow{\text{Scalar Poisson's equation for electric potential}} A_x = \frac{\mu_0}{4\pi} \oint_{u'} \frac{J_x}{r} du'$$

Scalar Poisson's equation for electric potential

$$\vec{\nabla}^2 f = -\frac{\rho}{\epsilon_0} \Rightarrow f = \frac{1}{4\pi\epsilon_0} \int_{u'} \frac{\rho}{r} du'$$

Combining the three components, we have

$$\boxed{\mathbf{A} = \frac{\mu_0}{4\pi} \int_{V'} \frac{\mathbf{J}}{R} dv' \quad (\text{Wb/m}).}$$



- **Magnetic Flux(磁通)**, Φ (through a given area S which is bounded by contour C):

$$\Phi = \int_S \mathbf{B} \cdot d\mathbf{s}.$$



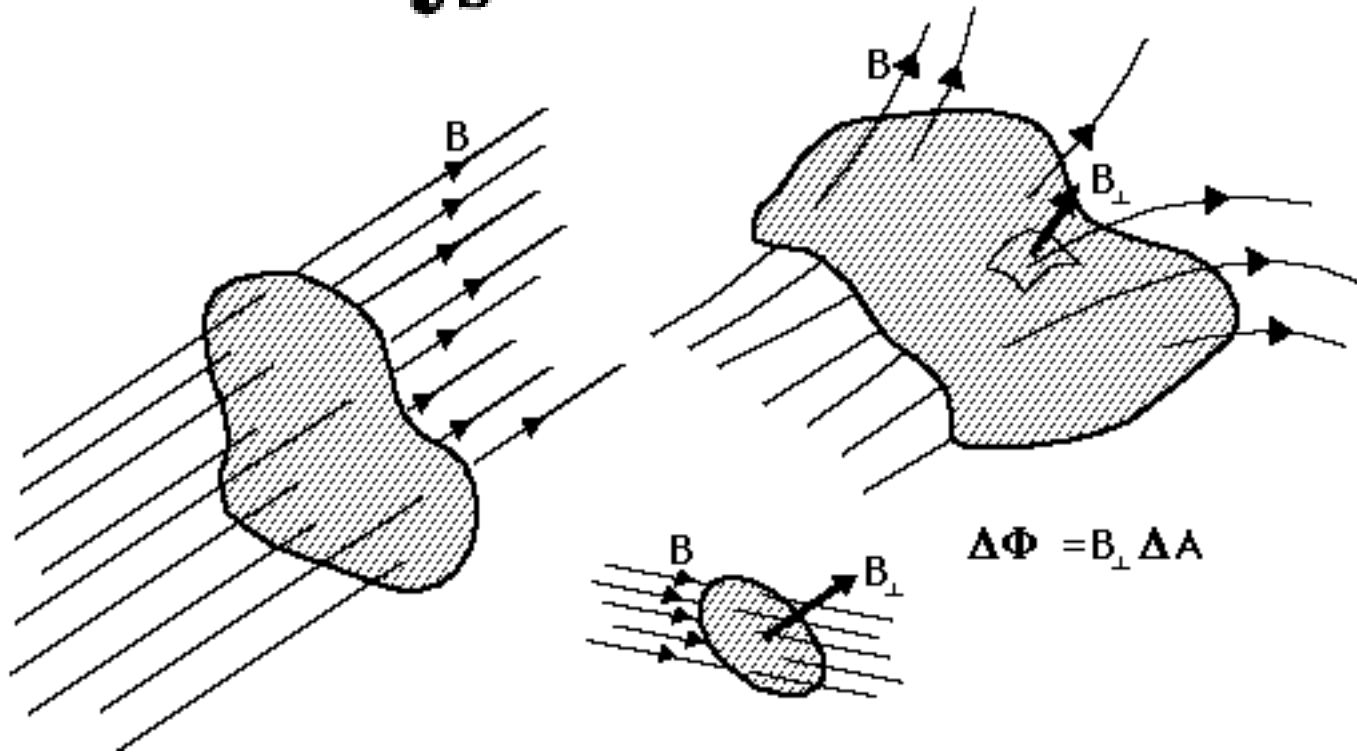
$$\vec{B} = \vec{\nabla} \times \vec{A}$$

$$\Phi = \int_S (\nabla \times \mathbf{A}) \cdot d\mathbf{s} = \oint_C \mathbf{A} \cdot d\boldsymbol{\ell}$$

Thus, vector magnetic potential $\mathbf{A}$ does have physical significance in that its line integral around any closed path equals the total magnetic flux passing through the area enclosed by the path.

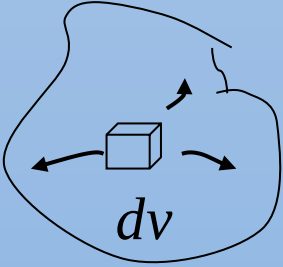
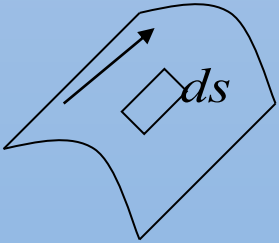
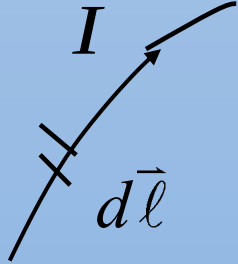
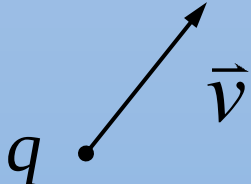
- **Magnetic Flux, Φ** (through a given area S which is bounded by contour C):

$$\Phi = \int_S \mathbf{B} \cdot d\mathbf{s}. \quad (\text{Web}), \text{ or } (\text{T} \cdot \text{m}^2)$$



6-4 Biot-Savart Law and applications

Magnetic field source

<u>3-dim</u> Volume current	<u>2-dim</u> Surface current	<u>1-dim</u> current	<u>0-dim</u> Moving charge
$\mathbf{J}$ (A/m ²)	$\mathbf{J}_s$ (A/m)	I (A)	q (C)
$\mathbf{J}d\mathbf{v}$ (A•m) 	$\mathbf{J}_s d\mathbf{s}$ (A•m) 	$I d\vec{\ell}$ (A•m) 	$q d\mathbf{v}$ (A•m) 

电流
微分
片段

For a thin wire with cross-sectional area S , dv' equals $S d\ell'$,

$$\mathbf{A} = \frac{\mu_0 I}{4\pi} \oint_{C'} \frac{d\ell'}{R} \quad \leftarrow \quad \mathbf{A} = \frac{\mu_0}{4\pi} \int_{V'} \frac{\mathbf{J}}{R} dv'$$

*A closed circuit that current flows (**static**)*

The magnetic flux density is then

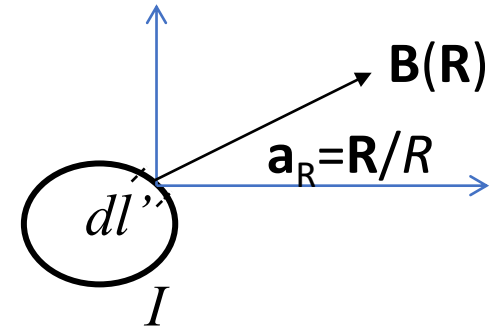
$$\begin{aligned} \mathbf{B} &= \nabla \times \mathbf{A} = \nabla \times \left[\frac{\mu_0 I}{4\pi} \oint_{C'} \frac{d\ell'}{R} \right] \\ &= \frac{\mu_0 I}{4\pi} \oint_{C'} \nabla \times \left(\frac{d\ell'}{R} \right). \end{aligned}$$

$$\nabla \times (f\mathbf{G}) = f\nabla \times \mathbf{G} + (\nabla f) \times \mathbf{G}.$$

$$= \frac{\mu_0 I}{4\pi} \oint_{C'} \left[\frac{1}{R} \underbrace{\nabla \times d\ell'}_{=0} + \left(\nabla \frac{1}{R} \right) \times d\ell' \right].$$

After the mathematical process (Eqs. 6.29-6.32) ..., we have

$$\mathbf{B} = \frac{\mu_0 I}{4\pi} \oint_{C'} \frac{d\boldsymbol{\ell}' \times \mathbf{a}_R}{R^2} \quad (\text{T}).$$



Biot-Savart Law(毕奥-萨伐尔定律): *Magnetic flux density caused by a current in a closed path C'*

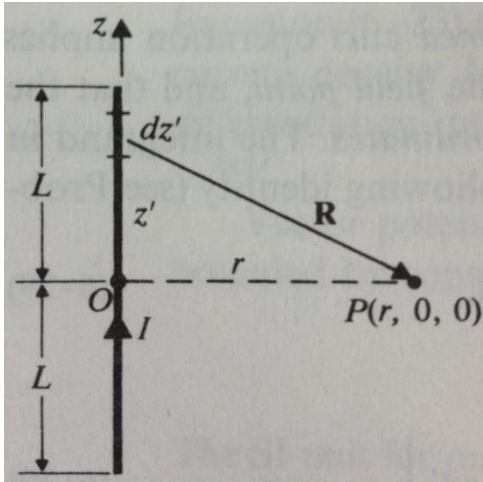
It can be expressed by two steps:

$$\mathbf{B} = \oint_{C'} d\mathbf{B} \quad \longleftarrow \quad d\mathbf{B} = \frac{\mu_0 I}{4\pi} \left(\frac{d\boldsymbol{\ell}' \times \mathbf{a}_R}{R^2} \right)$$

$d\mathbf{B}$: Magnetic flux density caused by a current element $I d\boldsymbol{\ell}$

$$\mathbf{a}_R = \frac{\mathbf{R}}{R} \quad \longrightarrow \quad d\mathbf{B} = \frac{\mu_0 I}{4\pi} \left(\frac{d\boldsymbol{\ell}' \times \mathbf{R}}{R^3} \right)$$

Q6-4 A direct current I flows in a straight wire of length $2L$. Find the magnetic flux density B at a point located at a distance r from the wire in the bisecting plane.



Method 1: find B from $\nabla \times \vec{A}$

$$d\vec{\ell} = \vec{a}_z dz'$$

The cylindrical coordinates of the field point P are $(r, 0, 0)$.

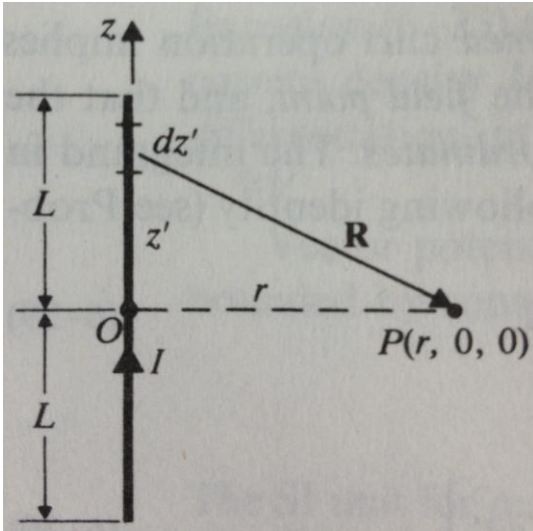
$$R = \sqrt{z'^2 + r^2}$$

$$\vec{A} = \frac{\mu_0 I}{4\pi} \oint \frac{d\vec{\ell}}{R} = \vec{a}_z \frac{\mu_0 I}{4\pi} \int_{-L}^L \frac{dz'}{\sqrt{z'^2 + r^2}} = \vec{a}_z \frac{\mu_0 I}{4\pi} \ln \frac{\sqrt{L^2 + r^2} + L}{\sqrt{L^2 + r^2} - L}$$

$$\vec{B} = \nabla \times \vec{A} = \nabla \times (\vec{a}_z A_z) = \vec{a}_r \frac{1}{r} \frac{\partial A_z}{\partial \phi} - \vec{a}_\phi \frac{\partial A_z}{\partial r}$$

Cylindrical symmetry around the wire assures that $\frac{\partial A_z}{\partial \phi} = 0$. Thus,

$$\vec{B} = -\vec{a}_\phi \frac{1}{r} \left[\frac{\mu_0 I}{4\pi} \ln \frac{\sqrt{L^2 + r^2} + L}{\sqrt{L^2 + r^2} - L} \right] = \vec{a}_\phi \frac{\mu_0 I L}{2\pi r \sqrt{L^2 + r^2}}.$$

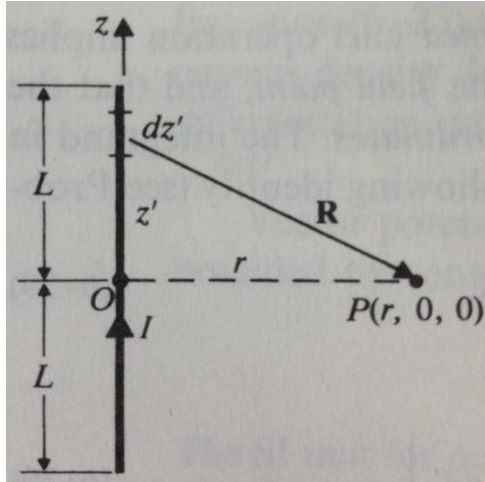


$$\vec{B} = -\vec{a}_f \frac{\mu_0 I}{4\pi r} \left[\frac{m_0 I}{4\pi} \ln \frac{\sqrt{L^2 + r^2} + L}{\sqrt{L^2 + r^2} - L} \right] = \vec{a}_f \frac{\mu_0 I L}{2\pi r \sqrt{L^2 + r^2}}.$$

When $r \ll L$, the above equation reduces to

$$\vec{B}_f = \vec{a}_f \frac{\mu_0 I}{2\pi r}$$

which is the expression for B at a point located at a distance r from an infinitely long, straight wire carrying current I



Method 2: applying Biot-Savart law

The distance vector from the source element to the field point P is

$$\vec{R} = \vec{a}_r r - \vec{a}_z z'$$

$$d\vec{B} = \frac{\mu_0 I}{4\pi} \left(\frac{d\vec{\ell}' \times \vec{R}}{R^3} \right)$$

$$d\ell' \vec{R} = \vec{a}_z dz' (\vec{a}_r r - \vec{a}_z z') = \vec{a}_r r dz'$$

$$\vec{B} = \oint d\vec{B} = \frac{\mu_0 I}{4\pi} \oint \frac{d\ell' \vec{R}}{R^3} = \vec{a}_r \frac{\mu_0 I}{4\pi} \int_{-L}^L \frac{r dz'}{(z'^2 + r^2)^{3/2}} = \vec{a}_r \frac{\mu_0 I L}{2\pi r \sqrt{L^2 + r^2}}$$

Ampere's circuital law:

Useful in determining B caused by a current I

there is a closed path C around the current such that the magnitude of B is constant over the path.

$$\oint_C \vec{B} \times d\vec{\ell} = \mu_0 I$$

Biot-Savart law:

Useful in determining B caused by a current I

there is no closed path C around the current such that the magnitude of B is constant over the path.

$$\mathbf{B} = \frac{\mu_0 I}{4\pi} \oint_{C'} \frac{d\ell' \times \mathbf{a}_R}{R^2} \quad (\text{T}).$$

- Static Electric Field in
free space/dielectric
(differential/integral form)

- Static Magnetic Field in
free space/dielectric
(differential/integral form)

- Steady Current density
(differential/integral form)

- **Static** Electric Field in
free space/dielectric
(differential/integral form)

$$\begin{aligned}\nabla \cdot \vec{E} &= \frac{\rho}{\epsilon_0} & \oint_S \vec{E} \cdot d\vec{s} &= \frac{Q}{\epsilon_0} \\ \nabla \times \vec{E} &= 0 & \oint_C \vec{E} \cdot d\vec{l} &= 0 \\ \nabla \cdot \vec{D} &= \rho & \oint_S \vec{D} \cdot d\vec{s} &= Q\end{aligned}$$

- **Steady** Current density
(differential/integral form)

- **Static** Magnetic Field in
free space/dielectric
(differential/integral form)

$\nabla \cdot \mathbf{B} = 0$	$\oint_S \mathbf{B} \cdot d\mathbf{s} = 0$
$\nabla \times \mathbf{B} = \mu_0 \mathbf{J}$	$\oint_C \mathbf{B} \cdot d\boldsymbol{\ell} = \mu_0 I$

$\nabla \cdot \mathbf{J} = 0$	$\oint_S \mathbf{J} \cdot d\mathbf{s} = 0$
$\nabla \times \left(\frac{\mathbf{J}}{\sigma} \right) = 0$	$\oint_C \frac{1}{\sigma} \mathbf{J} \cdot d\boldsymbol{\ell} = 0$

$$\mathbf{B} = \nabla \times \mathbf{A}$$

Ampere's circuital law:

Useful in determining $\mathbf{B}$ caused by a current I

there is a closed path C around the current such that the magnitude of B is constant over the path.

$$\oint_C \vec{B} \times d\vec{\ell} = \mu_0 I$$

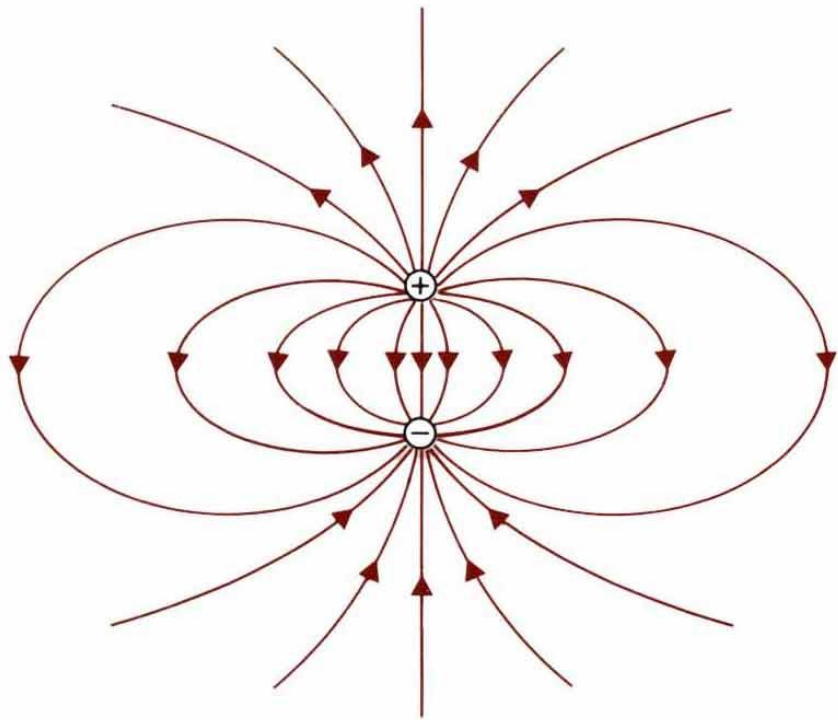
Biot-Savart law:

Useful in determining $\mathbf{B}$ caused by a current I

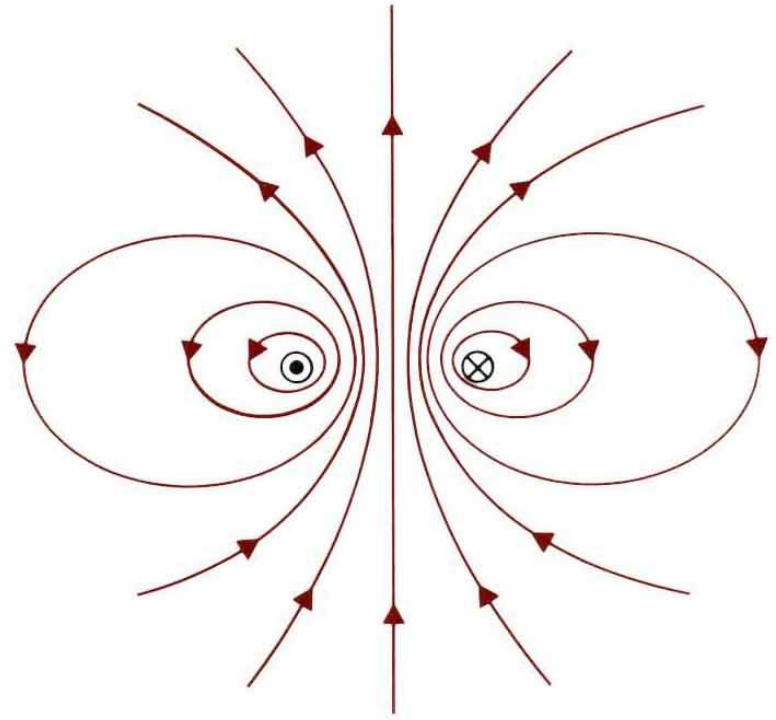
there is no closed path C around the current such that the magnitude of B is constant over the path.

$$\mathbf{B} = \frac{\mu_0 I}{4\pi} \oint_{C'} \frac{d\vec{\ell}' \times \mathbf{a}_R}{R^2} \quad (\text{T}).$$

6-5 Magnetic dipoles (磁偶极子)



(a) Electric dipole.



(b) Magnetic dipole.

FIGURE 6-9

Electric field lines of an electric dipole and magnetic flux lines of a magnetic dipole.

Calculate the magnetic flux density $\mathbf{B}$ of a magnetic dipole

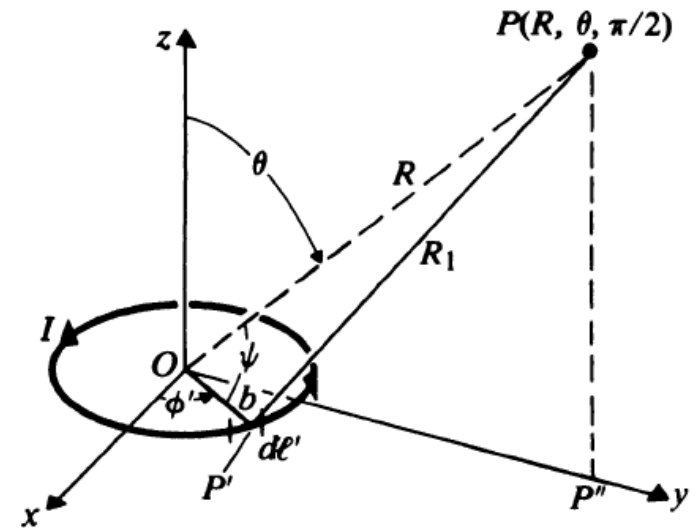
Method:

$$\mathbf{A} = \frac{\mu_0 I}{4\pi} \oint_{C'} \frac{d\ell'}{R_1} \xrightarrow{R^2 \gg b^2} \boxed{\mathbf{B} = \nabla \times \mathbf{A}}$$

Mathematical process (Eqs. 6.40-6.43) ..., we have

$$\mathbf{A} = \mathbf{a}_\phi \frac{\mu_0 I b^2}{4R^2} \sin \theta.$$

$$\mathbf{B} = \frac{\mu_0 I b^2}{4R^3} (\mathbf{a}_R 2 \cos \theta + \mathbf{a}_\theta \sin \theta),$$



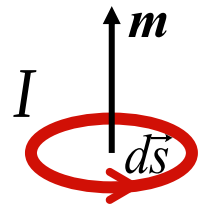
Magnetic dipole moment (磁偶极矩)

$$\mathbf{A} = \frac{\mu_0 \mathbf{m} \times \mathbf{a}_R}{4\pi R^2}$$



$$\mathbf{A} = \mathbf{a}_\phi \frac{\mu_0(I\pi b^2)}{4\pi R^2} \sin \theta$$

$$\mathbf{m} = \mathbf{a}_z I \pi b^2 = \mathbf{a}_z I S = \mathbf{a}_z m \quad (\text{A} \cdot \text{m}^2)$$



Magnetic dipole moment:

a vector = current $\times$ area of the loop; the direction of the thumb

Magnetic flux density of a magnetic dipole

$$\mathbf{B} = \frac{\mu_0 m}{4\pi R^3} (\mathbf{a}_R 2 \cos \theta + \mathbf{a}_\theta \sin \theta) \quad (\text{T})$$

Electric and magnetic dipoles

Electric dipole

$$\mathbf{p} = q\mathbf{d}.$$

$$V = \frac{\mathbf{p} \cdot \mathbf{a}_R}{4\pi\epsilon_0 R^2}$$

$$\mathbf{E} = \frac{p}{4\pi\epsilon_0 R^3} (\mathbf{a}_R 2 \cos \theta + \mathbf{a}_\theta \sin \theta).$$

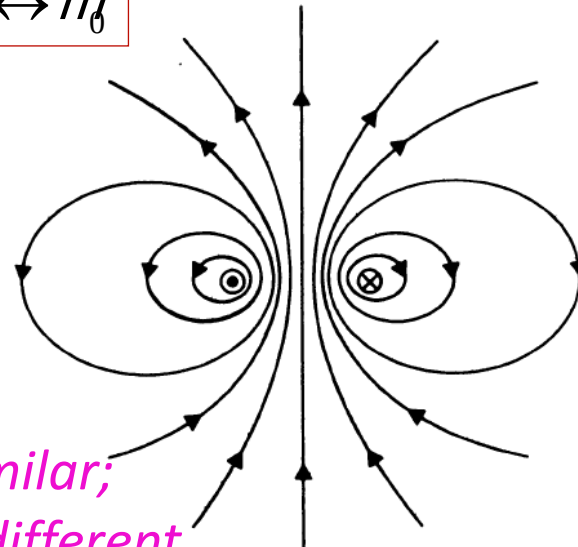
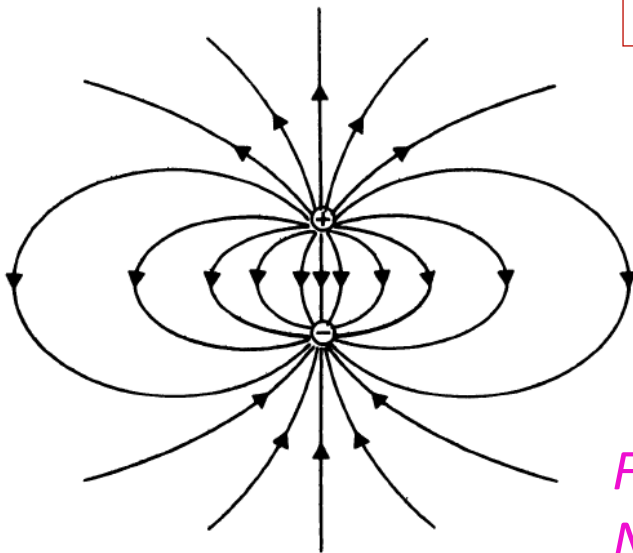
magnetic dipole

$$\mathbf{m} = \mathbf{a}_z I \pi b^2 = \mathbf{a}_z I S = \mathbf{a}_z m$$

$$\mathbf{A} = \frac{\mu_0 \mathbf{m} \times \mathbf{a}_R}{4\pi R^2}$$

$$\mathbf{B} = \frac{\mu_0 m}{4\pi R^3} (\mathbf{a}_R 2 \cos \theta + \mathbf{a}_\theta \sin \theta)$$

$$p \leftrightarrow m, \quad e_0^{-1} \leftrightarrow \mu_0$$



*Far-field is similar;
Near-field is different*

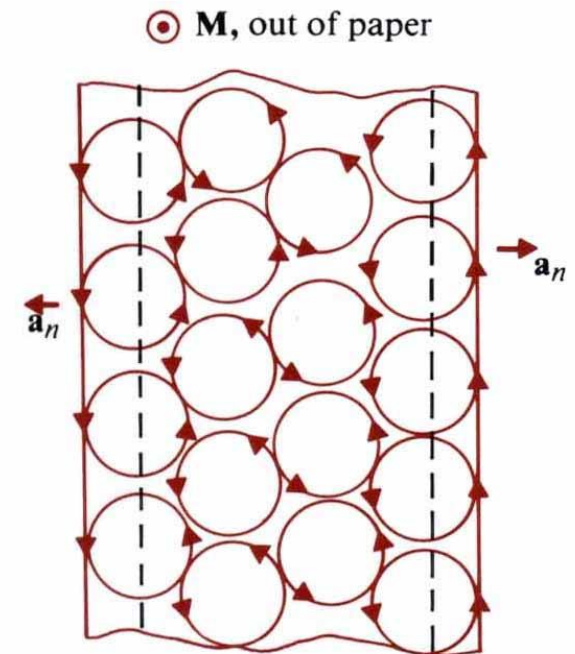
6-6 Magnetization and Equivalent Current Density

□ Magnetization vector (磁化强度矢量), $\mathbf{M}$

$$\mathbf{M} = \lim_{\Delta v \rightarrow 0} \frac{\sum_{k=1}^{n \Delta v} \mathbf{m}_k}{\Delta v} \quad (\text{A/m}),$$

The volume density of magnetic moment.

What will the postulated magnetic equations be in a magnetic medium?



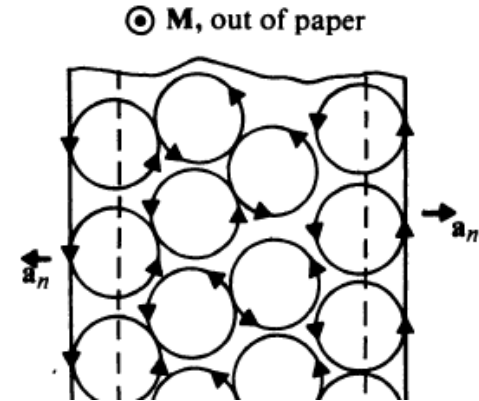
Vector magnetic potential due to a magnetic element dv'

$$d\mathbf{A} = \frac{\mu_0 \mathbf{M} \times \mathbf{a}_R}{4\pi R^2} dv'. \quad \leftarrow d\mathbf{m} = \mathbf{M} dv' \quad \boxed{\mathbf{A} = \frac{\mu_0 \mathbf{m} \times \mathbf{a}_R}{4\pi R^2}}$$

Derive A of a magnetic medium

$$\mathbf{A} = \int_{V'} d\mathbf{A} = \frac{\mu_0}{4\pi} \int_{V'} \mathbf{M} \times \nabla' \left(\frac{1}{R} \right) dv',$$

$$\mathbf{M} \times \nabla' \left(\frac{1}{R} \right) = \frac{1}{R} \nabla' \times \mathbf{M} - \nabla' \times \left(\frac{\mathbf{M}}{R} \right)$$



$$\mathbf{A} = \frac{\mu_0}{4\pi} \int_{V'} \frac{\nabla' \times \mathbf{M}}{R} dv' - \frac{\mu_0}{4\pi} \int_{V'} \nabla' \times \left(\frac{\mathbf{M}}{R} \right) dv'.$$

$$\int_{V'} \nabla' \times \mathbf{F} dv' = - \oint_{S'} \mathbf{F} \times d\mathbf{s}'$$

$$\mathbf{A} = \frac{\mu_0}{4\pi} \int_{V'} \frac{\nabla' \times \mathbf{M}}{R} dv' + \frac{\mu_0}{4\pi} \oint_{S'} \frac{\mathbf{M} \times \mathbf{a}'_n}{R} ds',$$

Equivalent volume and surface currents

$$\mathbf{A} = \frac{\mu_0}{4\pi} \int_{V'} \frac{\mathbf{J}}{R} dv' \quad (\text{Wb/m}).$$

$$\mathbf{A} = \frac{\mu_0}{4\pi} \int_{V'} \frac{\nabla' \times \mathbf{M}}{R} dv' + \frac{\mu_0}{4\pi} \oint_{S'} \frac{\mathbf{M} \times \mathbf{a}_n'}{R} ds',$$

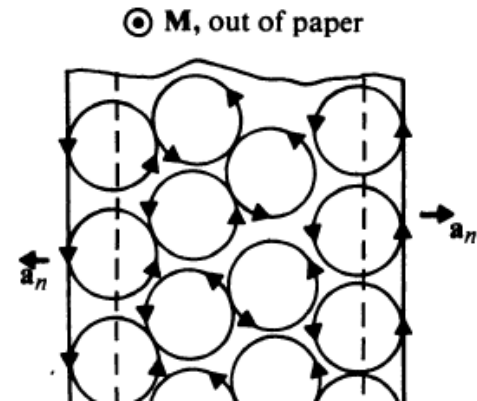
Magnetic current density

$$\mathbf{J}_m = \nabla \times \mathbf{M} \quad (\text{A/m}^2)$$

Equivalent volume current

$$\mathbf{J}_{ms} = \mathbf{M} \times \mathbf{a}_n \quad (\text{A/m}).$$

Equivalent surface current



The problem of finding the magnetic flux density $\mathbf{B}$ caused by a given volume density of magnetic dipole moment $\mathbf{M}$ is then reduced to finding the equivalent *magnetization current densities* $\mathbf{J}_m$ and $\mathbf{J}_{ms}$ by using Eqs. (6–62) and (6–63), determining $\mathbf{A}$ from Eq. (6–61), and then obtaining $\mathbf{B}$ from the curl of $\mathbf{A}$. The externally applied magnetic field, if it also exists, must be accounted for separately.

6-7 Magnetic Field Intensity and Relative Permeability

In the presence of magnetic materials, external magnetic field causes both internal dipole moments and induced magnetic moments, we can write

$$\frac{1}{\mu_0} \nabla \times \mathbf{B} = \mathbf{J} + \mathbf{J}_m = \mathbf{J} + \nabla \times \mathbf{M}$$

or

$$\nabla \times \left(\frac{\mathbf{B}}{\mu_0} - \mathbf{M} \right) = \mathbf{J}.$$

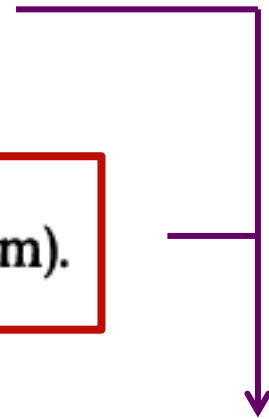
Magnetic field
intensity, $\mathbf{H}$

$$\mathbf{H} = \frac{\mathbf{B}}{\mu_0} - \mathbf{M} \quad (\text{A/m}).$$

Ampere's
circular theory

$$\oint_C \mathbf{H} \cdot d\boldsymbol{\ell} = I$$

$$\nabla \times \mathbf{H} = \mathbf{J}$$



Magnetic susceptibility and permeability

In linear and isotropic magnetic medium

$$\mathbf{M} = \chi_m \mathbf{H},$$

↑
Magnetic susceptibility (磁化率)

$$\begin{aligned}\mathbf{B} &= \mu_0(1 + \chi_m)\mathbf{H} \\ &= \mu_0\mu_r\mathbf{H} = \mu\mathbf{H} \quad (\text{Wb/m}^2)\end{aligned}$$

or

$$\mathbf{H} = \frac{1}{\mu} \mathbf{B} \quad (\text{A/m}),$$

Relative Permeability: $\mu_r = 1 + \chi_m = \frac{\mu}{\mu_0}$
(相对磁导率)

Electrostatics ↔ Magnetostatics	
$\mathbf{E}$	$\mathbf{B}$
$\mathbf{D}$	$\mathbf{H}$
ϵ	$\frac{1}{\mu}$
$\mathbf{P}$	$-\mathbf{M}$
ρ	$\mathbf{J}$
V	$\mathbf{A}$
$\cdot$	$\times$
$\times$	$\cdot$

Permeability of most materials is close to that of free space; for magnetic materials (Fe, Ni, Co, etc.), it could be 50-5000 or larger but depends on external field H .

6-9 Behavior of Magnetic Materials

$$\vec{M} = \chi_m \vec{H}, \chi_m: \text{magnetic susceptibility}$$
$$\vec{H} = \frac{1}{\mu} \vec{B}, \mu_r = 1 + \chi_m = \frac{\mu}{\mu_0}$$

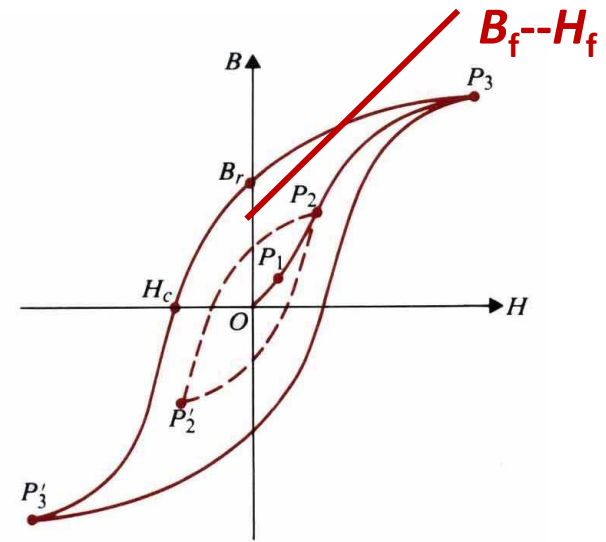


FIGURE 6-17

Hysteresis loops in the B - H plane for ferromagnetic material.

◎Diamagnetic反磁: $\mu_r \leq 1$ (χ_m :small negative number)

◎Paramagnetic顺磁: $\mu_r \geq 1$ (χ_m :small positive number)

◎Ferromagnetic铁磁: $\mu_r \gg 1$ (χ_m :large positive number)

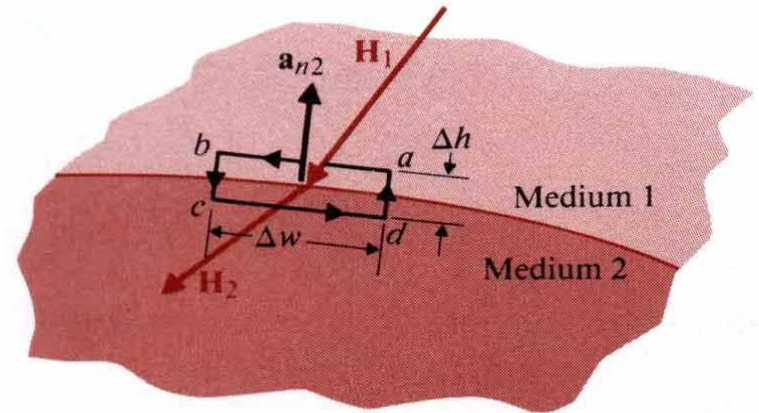
6-10 Boundary Conditions for Magnetostatic Field

■ Normal component

$$B_{1n} = B_{2n} \quad \leftarrow \quad \nabla \cdot \mathbf{B} = 0,$$

$$\mu_1 H_{1n} = \mu_2 H_{2n}.$$

The normal component of the magnetic flux density is continuous across an interface.

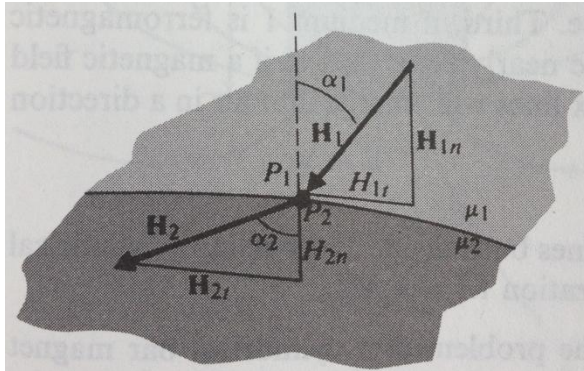


■ Tangent component

$$\mathbf{a}_{n2} \times (\mathbf{H}_1 - \mathbf{H}_2) = \mathbf{J}_s \quad \leftarrow \quad \nabla \times \mathbf{H} = \mathbf{J}.$$

The tangent component of the $\mathbf{H}$ field is discontinuous across an interface where a free surface current exists.

Q6-12 Two magnetic media with permeabilities μ_1 and μ_2 have a common boundary. The magnetic field intensity in medium 1 at the point P_1 has a magnitude H_1 and makes an angle α_1 with the normal. Determine the magnitude and the direction of the magnetic field intensity at point P_2 in medium 2.



Continuity of the normal component of B field

$$\mu_2 H_2 \cos \alpha_2 = \mu_1 H_1 \cos \alpha_1$$

Since neither of the media is a perfect conductor, the tangential component of H field is continuous

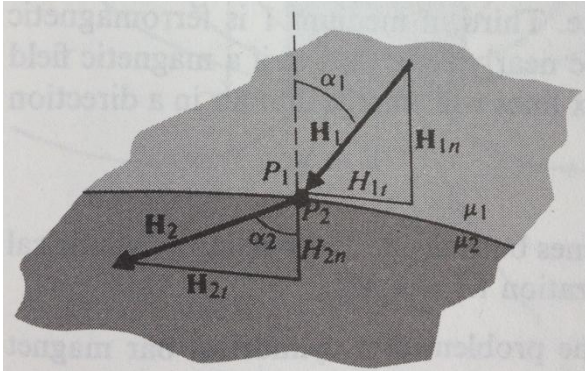
$$H_2 \sin \alpha_2 = H_1 \sin \alpha_1$$

$$\frac{\tan \alpha_2}{\tan \alpha_1} = \frac{\mu_2}{\mu_1}$$

The magnitude of $\vec{H}_2$ is

$$\begin{aligned} H_2 &= \sqrt{H_{2t}^2 + H_{2n}^2} = \sqrt{(H_2 \sin \alpha_2)^2 + (H_2 \cos \alpha_2)^2} \\ &= H_1 \left[\sin^2 \alpha_1 + \left(\frac{\mu_1}{\mu_2} \cos \alpha_1 \right)^2 \right]^{1/2} \end{aligned}$$

$$\frac{\tan \alpha_2}{\tan \alpha_1} = \frac{\mu_2}{\mu_1}$$



Discussion

(1) Medium 1: air; medium 2: iron

For any arbitrary angle α_1 that is not close to zero, the magnetic field in a ferromagnetic medium runs almost parallel to the interface.

(2) Medium 1: iron; medium 2: air

If a magnetic field originates in a ferromagnetic medium, the flux lines will emerge into air in a direction almost normal to the interface.

Magnetic materials: $\frac{\tan \alpha_2}{\tan \alpha_1} = \frac{\mu_2}{\mu_1}$

$$H_2 = H_1 \left[\sin^2 \alpha_1 + \left(\frac{\mu_1}{\mu_2} \cos \alpha_1 \right)^2 \right]^{1/2}$$

Conductors: $\frac{\tan \alpha_2}{\tan \alpha_1} = \frac{\sigma_2}{\sigma_1}$

$$J_2 = J_1 \left[\left(\frac{\sigma_2}{\sigma_1} \sin \alpha_1 \right)^2 + \cos^2 \alpha_1 \right]^{1/2}$$

Dielectric materials: $\frac{\tan \alpha_2}{\tan \alpha_1} = \frac{\epsilon_2}{\epsilon_1}$

$$E_2 = E_1 \left[\sin^2 \alpha_1 + \left(\frac{\epsilon_1}{\epsilon_2} \cos \alpha_1 \right)^2 \right]^{1/2}$$

6-12 Magnetic Energy

Magnetic energy

$$W_m = \frac{1}{2} \oint_{\mathcal{V}} (\vec{H} \times \vec{B}) dv$$

$$\vec{H} = \frac{\vec{B}}{m}$$

$$W_m = \frac{1}{2} \oint_{\mathcal{V}} \frac{B^2}{m} dv$$

$$W_m = \frac{1}{2} \oint_{\mathcal{V}} m H^2 dv$$

c.f. Electric energy

$$W_e = \frac{1}{2} \oint_{\mathcal{V}} (\vec{E} \times \vec{D}) dv$$

$$W_e = \frac{1}{2} \oint_{\mathcal{V}} e E^2 dv = \frac{1}{2} \oint_{\mathcal{V}} \frac{D^2}{e} dv$$